Digital Decimation Filter Design Challenge

Architecture and Coefficients Report

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1 Architecture Overview

The system utilizes a two-stage decimation architecture, achieving an overall Oversampling Ratio (OSR) of 2048. Direct single-rate FIR filtering at oversampled rates is computationally prohibitive. A practical decimation system uses a multiplier-free Cascaded Integrator-Comb (CIC) filter as Stage 1 to absorb the decimation factor, followed by a low-pass FIR compensator at a lower decimated clock.

1.1 Stage 1: CIC Decimation Filter

The first stage absorbs the primary decimation factor. To maintain exact mathematical correctness without intermediate clamping, registers must obey Hogenauer's bound. The final output is safely scaled and truncated to a maximum of 20 bits.

- **Filter Order (N):** 4
- **Differential Delay (M):** 2
- **Decimation Factor (R):** 512
- **Input Bus Width (B_{in}):** 2 bits
- **Internal Bus Width (B_{max}):** 42 bits

1.2 Stage 2: Low-Pass FIR Droop Compensator

CIC filters exhibit characteristic passband droop described by the sinc function response. A low-pass FIR filter is designed with passband and stopband edges pushed above 2000 to match the inverse-sinc profile, recovering high in-band tones (such as the 990 Hz test vector).

- **Architecture:** 253-stage Sequential Shift Register with Multiply-Accumulate (MAC) engine
- **Decimation Factor (R):** 4
- **Output Bus Limit:** 20 Bits

2 Performance Evaluation (PSD)

The primary evaluation metric is the Signal-to-Noise Ratio (SNR) calculated across specific test frequencies (10 Hz, 500 Hz, and 990 Hz). The Power Spectral Density (PSD) plots below illustrate the performance across all three benchmark tones.

The measured SNR figures from the test benches are **104.42 dB** for the 10 Hz tone, **160.28 dB** for the 500 Hz tone, and **101.37 dB** for the 990 Hz tone.

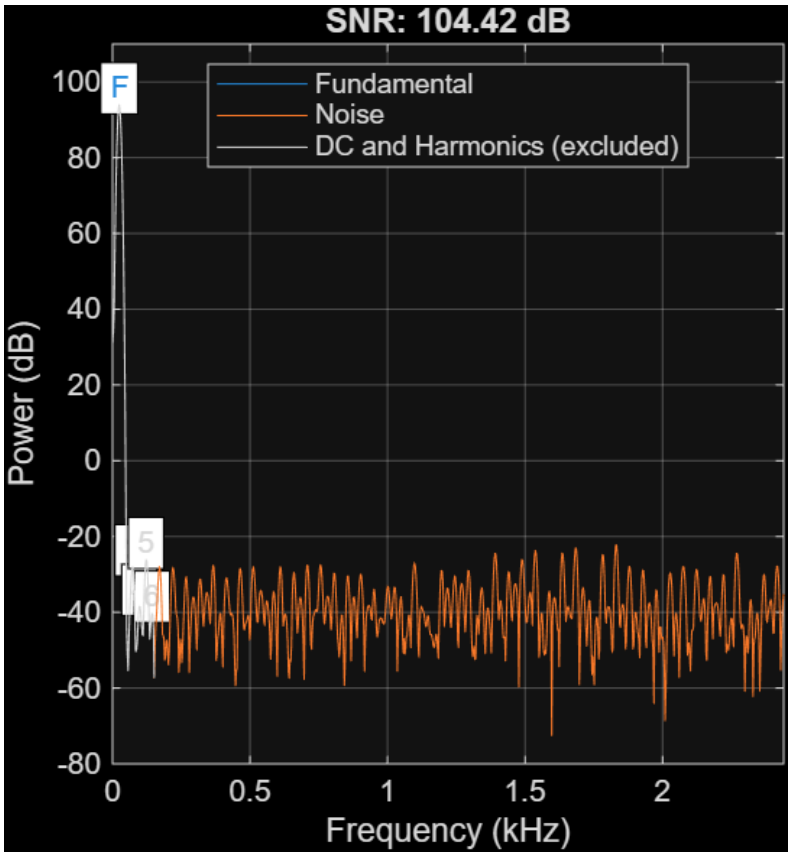


Figure 1: PSD Plot for 10 Hz Benchmark Tone (SNR: 104.42 dB)

3 FIR Filter Coefficients

The low-pass FIR compensator utilizes 253 taps. Due to even symmetry ($h[n] = h[252 - n]$), the coefficients are folded below.

Index	Tap Value	Index	Tap Value	Index	Tap Value
0, 252	4	42, 210	32	84, 168	-210
1, 251	5	43, 209	71	85, 167	-111
2, 250	-2	44, 208	63	86, 166	73
3, 249	-22	45, 207	10	87, 165	218
4, 248	-63	46, 206	-52	88, 164	217
5, 247	-118	47, 205	-80	89, 163	62
6, 246	-177	48, 204	-54	90, 162	-148

Index	Tap Value	Index	Tap Value	Index	Tap Value
7, 245	-218	49, 203	11	91, 161	-267
8, 244	-220	50, 202	72	92, 160	-204
9, 243	-174	51, 201	85	93, 159	9
10, 242	-90	52, 200	39	94, 158	232
11, 241	6	53, 199	-37	95, 157	306
12, 240	79	54, 198	-90	96, 156	167
13, 239	104	57, 195	64	99, 153	-327
14, 238	74	58, 194	104	100, 152	-98
15, 237	10	59, 193	73	101, 151	221
16, 236	-51	60, 192	-11	102, 150	411
17, 235	-78	61, 191	-92	103, 149	322
18, 234	-58	62, 190	-113	104, 148	-8
19, 233	-6	63, 189	-55	105, 147	-367
20, 232	46	64, 188	45	106, 146	-498
21, 231	67	65, 187	118	107, 145	-281
22, 230	45	66, 186	112	108, 144	167
23, 229	-4	67, 185	27	109, 143	548
24, 228	-49	68, 184	-82	110, 142	577
25, 227	-61	69, 183	-139	111, 141	184
26, 226	-33	70, 182	-101	112, 140	-403
27, 225	17	71, 181	11	113, 139	-784
28, 224	55	72, 180	121	114, 138	-644
29, 223	57	73, 179	152	115, 137	8
30, 222	20	74, 178	77	116, 136	787
31, 221	-30	75, 177	-56	117, 135	1138
32, 220	-61	76, 176	-158	118, 134	694
33, 219	-50	77, 175	-154	119, 133	-426
34, 218	-5	78, 174	-41	120, 132	-1580
35, 217	45	79, 173	108	121, 131	-1890
36, 216	65	80, 172	189	122, 130	-726
37, 215	41	81, 171	141	123, 129	1885
38, 214	-12	82, 170	-10	124, 128	5161
39, 213	-59	83, 169	-163	125, 127	7877
40, 212	-66				
41, 211	-28				
Center Tap					
126	8929				

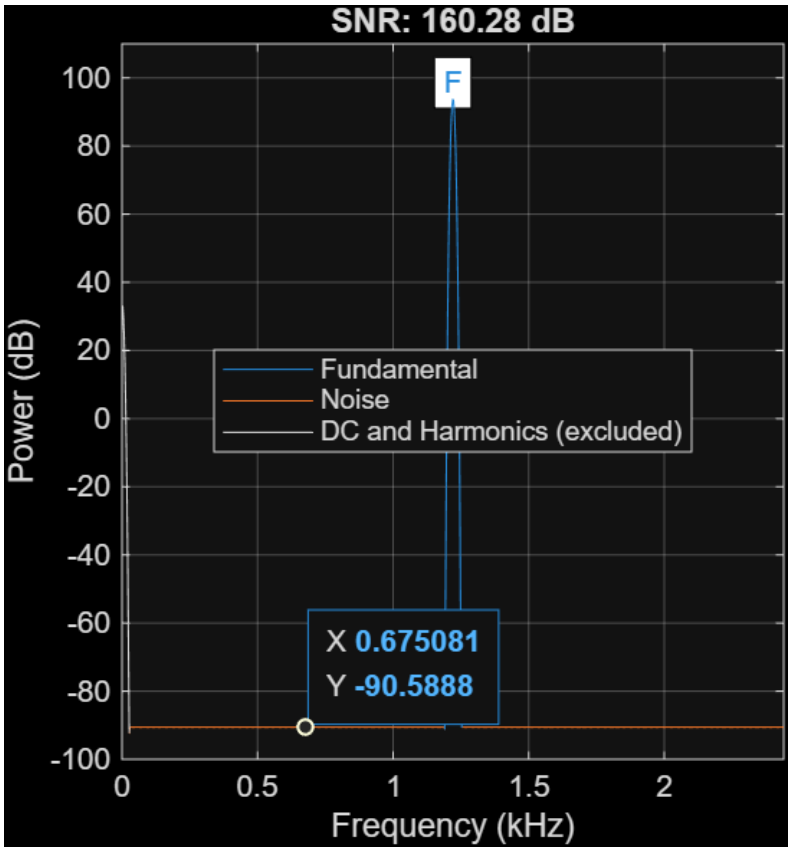


Figure 2: PSD Plot for 500 Hz Benchmark Tone (SNR: 160.28 dB)

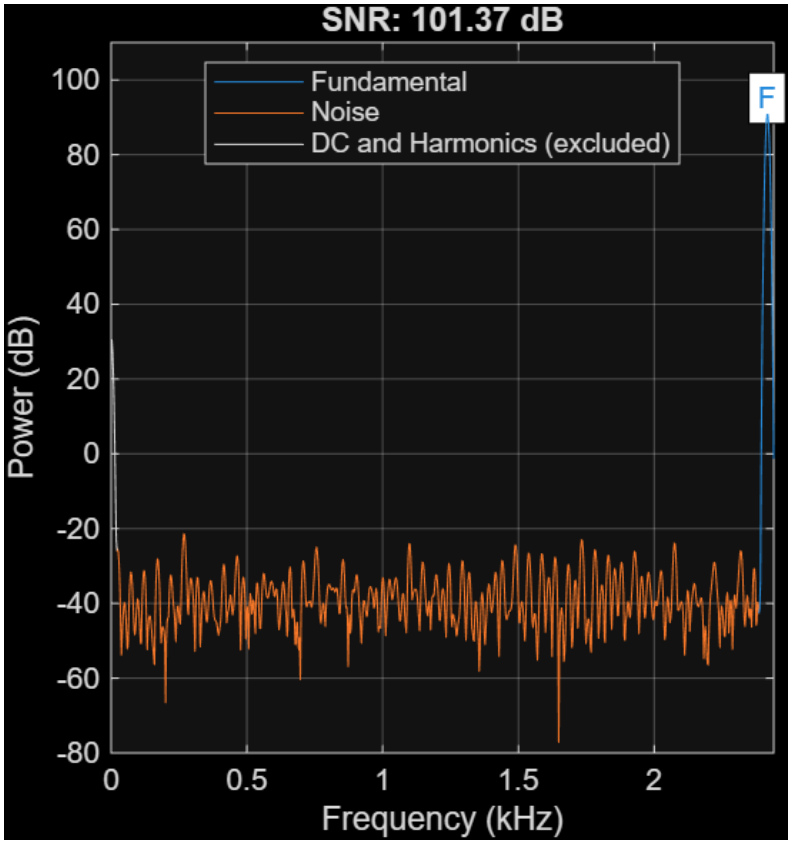


Figure 3: PSD Plot for 990 Hz Benchmark Tone (SNR: 101.37 dB)